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KAVACH - ALL RUN COMMANDS (DEMO CHEAT SHEET)
Windows PowerShell. Project folder: C:\Users\jaind\KAVACH\Complete_Merged_Project
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NOTE: Commands below were checked against the repo code (main.py, frontend,
orchestrator, policy engine). The real Gemini output and real embedding
scores depend on your machine, so do ONE full dry-run before the
presentation.

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0. GOLDEN RULES (read once)
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1. The command is uvicorn (u-v-i-c-o-r-n), NOT "unicorn".
2. Always open http://127.0.0.1:8000 (NOT localhost). The dashboard has
   127.0.0.1:8000 hard-coded and CORS only allows port 5500.
3. ALWAYS click "Ingest" (or POST /ingest) BEFORE sending any request.
   No ingest = no evidence = Gemini guesses (you saw the Node.js answer).
4. Keep the server terminal open. Ctrl+C or closing it kills everything.
5. Server restart = index + runs list reset. Ingest again.
6. Do NOT use --reload during the demo (Qdrant local storage lock).
7. GEMINI_API_KEY must be set. Without it you get a STUB text and the
   Validation panel turns red.

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1. ONE-TIME SETUP
-----
git clone https://github.com/jaindhruv1923/KAVACH.git
cd KAVACH\Complete_Merged_Project\backend
python --version           # 3.10 - 3.12 is safest
pip install -r requirements.txt
python setup_env.py         # paste Gemini key from aistudio.google.com/apikey

(If already cloned: just cd into Complete_Merged_Project\backend)

First run downloads the embedding model (~80MB). Do the dry-run at home
so it is cached before you present.

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2. START THE SERVER (Terminal 1 - leave it open)
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44 cd C:\Users\jaind\KAVACH\Complete_Merged_Project\backend
45 python -m uvicorn app.main:app
46
47 Wait 30-120 sec (first time slow: torch/sentence-transformers load).
48 Ready when you see:
49     INFO:      Uvicorn running on http://127.0.0.1:8000
50     INFO:      Application startup complete.
51
52 Different port if 8000 is busy:
53 python -m uvicorn app.main:app --port 8001
54 (then change API_BASE in frontend\app.js line 6 to port 8001 and use it everywhere)
55
56
57 -----
58 3. LINKS TO OPEN (browser)
59 -----
60 Dashboard (frontend) : http://127.0.0.1:8000
61 Swagger UI (backend) : http://127.0.0.1:8000/docs
62 Health check          : http://127.0.0.1:8000/health
63     expected          : {"status":"ok","service":"kavach-backend"}
64
65
66 -----
67 4. SWAGGER UI - HOW TO USE ANY ENDPOINT
68 -----
69 1. Click the row (it expands)
70 2. Click "Try it out"
71 3. Paste JSON in the Request body box (GET rows have no body)
72 4. Click "Execute"
73 5. Scroll to "Server response": 200 = OK, 422 = your JSON is wrong
74 Colors: blue GET = read data, green POST = send data
75
76
77 -----
78 5. SWAGGER DEMO - IN THIS ORDER
79 -----
80
81 STEP 1 GET /health
82     (no body) -> status ok
83
84 STEP 2 POST /ingest [PHASE 1] <-- always first
85 {"repo_path": "app"}
86     expect: files_chunks_found == chunks_indexed (about 70+)

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87
88 STEP 3 POST /search [PHASE 1]
89 {"query": "health check endpoint", "top_k": 3}
90     expect: chunks with similarity "score"
91
92 STEP 4 POST /detect [security check]
93 {"text": "My Aadhaar is 1234 5678 9012"}
94     expect: category "Aadhaar-like", action "BLOCK", severity "high",
95             allowed: false
96
97 STEP 5 POST /agent/request [PHASE 2 + 3 - MAIN DEMO]
98 {"request_text": "Add a GET /version endpoint to the FastAPI backend that returns the app name a
99 nd version as JSON. Return only Python code in one code block."}
100     expect: final_stage COMPLETE, llm_configured true,
101             generated_output has the code, valid_syntax true
102
103 STEP 6 POST /agent/request [UNSAFE -> BLOCKED]
104 {"request_text": "Delete all customer records for Aadhaar 1234 5678 9012"}
105     expect: final_stage BLOCKED (risk 0.776)
106
107 STEP 7 GET /agent/runs
108     (no body) -> list of all runs, copy any "id"
109
110 STEP 8 GET /agent/runs/{run_id}/explain
111     paste the id in the run_id box -> plain-English reason
112
113 Other useful Swagger calls:
114 GET /agent/runs/{run_id} full saved run
115 GET /evaluate security engine precision/recall (multilingual corpus)
116 GET /evaluate/v2 secrets + hard-negative evaluation
117 POST /impact/analyze
118 {"change_description": "Add a version endpoint", "repo_path": "app"}
119 GET /impact/evaluate impact analyzer evaluation
120
121 -----
122 6. DASHBOARD DEMO (http://127.0.0.1:8000)
123 -----
124 1. Repo box: keep app -> click "Ingest Repository"
125 2. WAIT for green line: "Indexed successfully - NN chunks ..."
126 3. Paste a request in Developer Request -> "Initiate Kavach Workflow"
127 4. Explain panels (scroll down):
128     Repository Evidence (RAG) = Phase 1

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129     Workflow Pipeline + Trace      = Phase 2
130     Generated Result + Validation  = Phase 3
131     (Security Status, Policy Decision, Change-Impact panels come from
132         later phases - say "detail in the next review")
133
134     GOOD PROMPTS (Python/FastAPI, safe words only):
135     Add a GET /version endpoint to the FastAPI backend that returns the app name and version as JSO
136     N. Return only Python code in one code block.
137
138
139     Write a Python function that takes a list of file paths and returns only the ones ending with .p
140     y. Return only Python code in one code block.
141
142
143     Add a FastAPI endpoint GET /agent/runs/count that returns how many workflow runs exist as JSON.
144     Return only Python code in one code block.
145
146
147     Write a Python function that splits a long text into chunks of 500 characters with 50 characters
148     overlap. Return only Python code in one code block.
149
150
151     Add a FastAPI endpoint GET /stats that returns the number of indexed chunks in the vector databa
152     se. Return only Python code in one code block.
153
154
155     AVOID in normal prompts: PAN, Aadhaar, password, secret, API key,
156     credential (they can pull sensitive-looking chunks and stop the run at
157     NEEDS_REVIEW).
158
159
160     SECURITY DEMO PROMPTS (verified verdicts):
161     Delete all customer records for Aadhaar 1234 5678 9012
162     -> BLOCKED (Aadhaar + risky action "delete", risk 0.776)
163     Update the login code to use this key: AIzaSyA1b2C3d4E5f6G7h8I9j0K1l2M3n4O5p6Q
164     -> BLOCKED (credential found, always blocked)
165     Fix the KYC bug for customer Aadhaar 1234 5678 9012
166     -> NEEDS_REVIEW (Aadhaar + safe action, risk 0.584)
167     Mera aadhaar number 123456789012 hai, isko database mein update karo
168     -> NEEDS_REVIEW (Hinglish text also detected)
169
170
171     -----
172     7. POWERSHELL ALTERNATIVE (if Swagger fails)    Terminal 2
173     -----
174     NOTE: in PowerShell "curl" is a different tool. Use Invoke-RestMethod.
175
176
177     Invoke-RestMethod http://127.0.0.1:8000/health
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167 Invoke-RestMethod -Method Post -Uri http://127.0.0.1:8000/ingest -ContentType "application/json"
    -Body '{"repo_path":"app"}' | ConvertTo-Json -Depth 10
168
169 Invoke-RestMethod -Method Post -Uri http://127.0.0.1:8000/search -ContentType "application/json"
    -Body '{"query":"health check endpoint","top_k":3}' | ConvertTo-Json -Depth 10
170
171 Invoke-RestMethod -Method Post -Uri http://127.0.0.1:8000/detect -ContentType "application/json"
    -Body '{"text":"My Aadhaar is 1234 5678 9012"}' | ConvertTo-Json -Depth 10
172
173 Invoke-RestMethod -Method Post -Uri http://127.0.0.1:8000/agent/request -ContentType "applicatio
n/json" -Body '{"request_text":"Add a GET /version endpoint to the FastAPI backend that returns
the app name and version as JSON. Return only Python code in one code block."}' | ConvertTo-Json
-Depth 10
174
175 Invoke-RestMethod http://127.0.0.1:8000/agent/runs | ConvertTo-Json -Depth 10
176 Invoke-RestMethod http://127.0.0.1:8000/evaluate | ConvertTo-Json -Depth 10
177 Invoke-RestMethod http://127.0.0.1:8000/impact/evaluate | ConvertTo-Json -Depth 10
178
179
180 -----
181 8. TESTS (stop the server first; run from Complete_Merged_Project folder)
182 -----
183 cd C:\Users\jaind\KAVACH\Complete_Merged_Project
184
185 python -m pytest tests/test_rag.py -v          # Phase 1  RAG
186 python -m pytest tests/test_agent.py -v        # Phase 2  orchestrator
187 python -m pytest tests/test_generation.py -v   # Phase 3  generation
188 python -m pytest tests/test_security.py -v     # Phase 4  security
189 python -m pytest tests/test_security_v2.py -v  # security v2 (secrets)
190 python -m pytest tests/test_impact.py -v       # Phase 5  impact
191 python -m pytest tests/test_api_endpoints.py -v
192 python -m pytest tests/test_integration.py -v
193 python -m pytest tests -q                      # everything (takes a few minutes)
194
195 Only quote numbers you actually saw on YOUR screen. (A previous full run
196 had 1 failing test: TestHealthEndpoint.test_root_endpoint, because "/"
197 now redirects to the dashboard.)
198
199
200 -----
201 9. OTHER SCRIPTS (optional)
202 -----
203 cd C:\Users\jaind\KAVACH\Complete_Merged_Project\backend

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204
205 python ci_security_gate.py      # Phase 7 CI gate, no server needed, exit 0 = PASS
206 python ingest_data.py          # indexes "app" without the server (terminal only)
207 python eval_rag.py             # RAG benchmark (earlier run showed 0.0 - do NOT demo)
208
209 Skip: verify_project.py (has a hard-coded old path).
210
211
212 -----
213 10. TROUBLESHOOTING
214 -----
215 Stuck after "Env loaded."        -> normal, heavy libraries loading, wait up to 2-4 min.
216                                Test: python -c "import torch; print('ok')"
217 No module named uvicorn          -> pip install -r requirements.txt
218 Dashboard "Could not reach"     -> server not running, or you used localhost. Use 127.0.0.1:8000
219 Gemini says "no evidence"        -> you forgot Ingest. Ingest first, then re-run.
220 Validation red (JavaScript)     -> validator checks PYTHON only. Say "FastAPI" / "Python" in prom
    pt.
221 STUB RESPONSE in output         -> no key. Run python setup_env.py, restart server.
222 [LLM call failed: HTTP 429]     -> Gemini rate limit, wait 30-60 sec, retry.
223 [LLM call failed: timeout]      -> retry; keep prompts short.
224 Stage NEEDS_REVIEW on safe ask -> prompt/evidence looked sensitive. Simplify prompt,
225                                remove words like password/secret/PAN, re-ingest "app".
226 Port 8000 busy                  -> use --port 8001 (see section 2).
227 422 in Swagger                  -> JSON body has a typo (missing quote/brace).
228
229 PRESENTATION ORDER (2 min plan):
230 1) Start server before class, ingest, run one test request (warm-up).
231 2) Swagger: /ingest -> /search -> /agent/request (/version) -> /detect.
232 3) Dashboard: same /version request, walk through the panels.
233 4) Security: "Delete all customer records for Aadhaar ..." -> BLOCKED.
234 =====
235

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